Rishav Roy

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Education

University of Wisconsin–Madison, B.S. Spring 2026
Triple Major: Economics, Mathematics, and Data Science *Madison, WI*
Recent GPA (since Summer 2024): 3.9 | Overall GPA: 3.3

Graduate-Level Coursework

- PhD Macroeconomics I–II (MATLAB)
  Top scorer in PhD Macroeconomics II
- PhD Microeconomics I–II
- PhD Industrial Organization I–II (audit)
- PhD Econometrics I–II (Stata)
  Top scorer in PhD Econometrics II in 18-cr. semester
- Master's-Level Measure & Integration (Honors)
  Top scorer of first half in 18-cr. semester

Select Undergraduate Coursework

- Data Science Programming I–II (Python, SQL)
- Introduction to Optimization (Julia)
- Machine Learning & Classification (Python)
- Econometrics: AI & Machine Learning (R)
- Foundation Models & LLMs (Python)
- Real Analysis I–II (Honors)

Research Experience

Escaping Inequality in India: The Role of English-Medium Instruction Spring 2025–Present
Senior Thesis Paper (with feedback from Prof. Laura Schechter)

- Built a district pseudo-panel ($N = 454$) merging 2007-08/2017-18 survey microdata and metadata, 35 Census 2001 language files, & 2019-20 shapefiles to study EMI exposure on consumption growth with survey-weighted measures.
- Estimated district-level 2SLS with state-level clustered SEs, partial $F = 39.20$. Documented endogeneity threats, spatial spillovers (Moran's I, Monte Carlo). Designed reusable IV generator to safely iterate 2SLS specifications.
- Estimated enrollment probit with design-based SEs, complex survey design (nested strata, lonely PSUs). Computed AME, delta-method SEs via autodiff. Rejected MCAR via logits, BH procedure. Mitigated systematic missingness.
- Harmonized districts across renames/mergers/splits & typos via multi-pass fuzzy joins (soundex, q-gram, etc.), canonicalization, & 80 manual matches. Engineered data intake tools robust to OS-specific file-naming conventions.
- Drafted 49-page paper in R Markdown with detailed replication notes, automated publication-quality maps/tables.

Language Models as Measurement Instruments: Risks for Estimation Spring 2026–Present
Independent Research Project (with feedback from Prof. Harold D. Chiang)

- Formulated LM-generated measures as noisy proxies for central banks' latent policy stance. Decomposed noise into error/leakage channels whose effects on downstream economic estimation I then experimentally isolated.
- Built reproducible Python codebase for World Central Banks corpus (380,200 sentences, 3,453 docs, 1996-2024): ingestion, measurement, aggregation, FE regressions with IMF/World Bank data, and automated tables/figures.
- Evaluated non-classical measurement error using 25,000 human-labeled sentences and entity masking, semantic perturbations, validation sample-size simulations, regressions predicting errors, and sensitivity to aggregation rules.
- Currently fine-tuning RoBERTa model while using a TF-IDF/logit baseline to benchmark and debug the pipeline. Will compare models by predictive performance, potential propagation of errors into macroeconomic estimation.

Fast Inference for Large-N Panel Quantile Regression & Common Shocks Spring 2026–Present
Research Project (with guidance from Prof. Harold D. Chiang)

- Defined stochastic subgradient estimator for fixed-effects panel quantile regression under common shocks. Demonstrated $\sqrt{T}$-normality under full-period stochastic increments. Outlined Polyak-Ruppert averaging, online updating, and mini-batch rate conditions for $o_p(T^{-1/2})$ score residuals and stochastic approximation errors.
- Eliminated first-order effect of FE scores and N-dimensional incidental parameters via orthogonalized slope moment, yielding the Schur-complement Jacobian Γ and identifying the common-shock variance component Σ.
- Developed proof strategy for first-order equivalence between one-step corrected SGD estimator and exact FEQR: asymptotically negligible KKT residuals imply $\sqrt{T}$-normality and valid $\Gamma^{-1}\Sigma\Gamma^{-1}$ sandwich inference.

Empirical Implications of Stochastic Hysteresis on Optimal Policy Fall 2025
Independent Research Project (with guidance from Prof. Job Boerma)

- Derived observable proxies for Boerma et al. (2023)'s Malliavin/Itô calculus given CRRA utilities, CRS, etc. to distinguish true hysteresis from persistence or long memory in macro, finance, and climate policy.

- Developed framework to falsify large class of Markov models (e.g., DSGE, VAR) using non-Markov time series. Assessed computational and identification tradeoffs of Kalman filtering, particle filtering, and Bayesian MCMC.

Dynamic Discrete Choice in Fast Food: Loyalty, Technology, & Inequality Summer–Fall 2025
Independent Research Project (with feedback from Prof. Amanda Smith)

- Formulated and numerically optimized a structural model of discounts, channel choice (in-store vs. app vs. outside option), and loyalty adoption using exponential cone optimization (Julia/JuMP, MOSEK) with MNL demand.
- Built a reproducible simulation procedure (e.g., $T = 20$) with evolving states, feasibility/termination checks, & numerical stability via stable softmax. Outlined counterfactuals for COVID, input-cost inflation, inequality shocks.
- Ensured convexity despite MNL demand & concave technology investment via log-sum-exp & RSOC epigraphs. Avoided bilinearity by updating loyalty shares via lagged states. Added income segmentation via MILP Big-M.

Work Experience

Economic Consultant and Data Analyst: Capital and Firm Ownership in Retirement 2024
Maui Mastermind Business Coaching *Remote*

- Harmonized SCF 2022 microdata with automated data integrity checks to characterize role of business ownership in post-retirement wealth persistence in R. Implemented survey design with 5 implicates, 999 replicate weights.
- Estimated age-grouped wealth distributions via survey-weighted quantiles (P50-P99.99) with replicate-weight SEs combined under multiple imputation. Illustrated and exported results in client-readable tables/figures.
- Produced breakdowns of asset/income composition (e.g., active/non-active business ownership) via nearest-neighbor selection by net worth for age×percentile estimates. Detailed SCF measurement/sampling limitations, data gaps.

Restaurant and Inventory Manager 2022–2023
Sookie's Veggie Burgers *Madison, WI*

- Built data-driven inventory tracker, cutting waste/holding times and building teamwork under stressful conditions.

Tutor of College-Level Economics, Calculus, Chemistry, and US History 2022–2023
Self-Employed *Remote*

- Improved student performance using self-made problem sets and exams, clear intuition, and engaging exposition.

Extracurricular Experience

Policies for Labor Recovery: Economic Development Case Competition (*1st Place*) Fall 2024
Sole Author of Written Report & Collaborator on Presentation (with feedback from Prof. Simeon Alder)

- Analyzed local general equilibrium impacts of a coal-plant closure using demand spillovers, labor reallocation, BLS and WI DWD projections, and direct/indirect/induced labor/fiscal losses ($251M output, $2.8M taxes, 308 jobs).
- Designed demand-driven training addressing job-specific human capital, participation constraints (school funding, housing supply), and retention via supportive services (childcare, transportation) & post-training mentorship guarantees. Evaluated funding constraints and alternatives beyond WIOA (e.g., WI Fast Forward, SCSEP).
- Presented strategies and potential partnerships for improving business dynamism and climate (licensing, zoning, knowledge diffusion) given local demand gaps (e.g., 50% retail furniture leakage) to Alliant Energy, MadREP.

Presenter, PREDOC.org Research Conference 2026
Booth School of Business, The University of Chicago

Presenter, Undergraduate Research Symposium 2025
University of Wisconsin-Madison

Awards & Honors

Veldor Kopitzke Scholarship in Economics Fall 2025
Department of Economics, University of Wisconsin-Madison

Dean's List Fall 2023, Spring 2025, Spring 2026
University of Wisconsin-Madison

Leadership Recognition: President's Volunteer Service Award Winner (Field Museum),
Martin Luther King Jr. Outstanding Young Person Award (Urban League of Greater Madison),
Certificate of Distinguished Achievement for Outstanding Citizenship (American Legion)

Technical Skills

Proficient: R (incl. spdep), Python (incl. sklearn, transformers), Stata, Julia, MATLAB, SQL, Git, LaTeX
Familiar: C++, Fortran, JavaScript, HTML/CSS, Mathematica